Product Rigidity of Marked Periodic Nerves for Unequal Simplex-Strand Completion Fibers

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Abstract

For a finite lattice configuration $P \subset L$, consider the periodic simplicial complex whose faces are finite sets $F \subset L$ satisfying $c - F \subseteq P$ for some $c \in L$, and its translation quotient $Q(P)$. We examine families supported on independent transverse vertices of a unimodular simplex, with an arbitrary nondegenerate interval of lattice points above each vertex. Although unequal interval lengths prevent such a configuration from being a literal Cartesian product, we prove equivalence under continuous deformation

$$Q(P) \simeq \left(\bigvee_{j=1}^{s} S^1 \right) \times S^1$$

for all transverse dimensions $s \geq 1$. The main technical point is an exact decomposition of the *full* inverse image of each transverse face into a mixed-strand core and horizontal thickenings. This allows a functorial facewise contraction and a rigorous descent through both translation groups. We compute the integral singular groups and the marked fundamental group, and apply the result to all shortest-completion fibers along an infinite target family of one fixed six-action, residual-rank-two integer source. The family exhibits a single change in topological type followed by rigidity even while optimal multiplicity grows without bound. The construction distinguishes marked periodic quotients from toric indispensable complexes and provides a uniform topology formula beyond literal Cartesian-product fibers.

Keywords. shortest completion; lattice-convex fiber; periodic simplicial nerve; marked displacement; cellular face diagram; vector partition function.

1 Introduction and scope

The shortest-completion geometry of integer counting sources arises from the setting of [1]. Fix a matrix $A \in \mathbb{Z}^{d \times n}$ and residual $r \in \mathbb{Z}^d$. Its minimal length and complete set of minimizers are

$$L_A(r) = \min \left\{ \sum_{i=1}^{n} z_i : z \in \mathbb{N}^n,\ Az = r \right\}, \qquad P_A(r) = \left\{ z \in \mathbb{N}^n : Az = r,\ \sum_i z_i = L_A(r) \right\}. \tag{1}$$

The set $P_A(r)$, rather than a selected minimizer, is intrinsic to the source. It is finite whenever the minimum exists; after translation, its difference lattice and the full fiber determine a periodic simplicial quotient. The periodic quotient and its compression from shortest-prefix geometry are developed in [2]; minimal-fiber and Cartesian-product laws in

[3]; and displacement marking in [4]. These existing results provide the baseline for the unequal-strand theorem.

Our object is the following parameterized family. For $s \geq 1$ choose the vertices $q_0 = 0, q_j = e_j$ of the unimodular simplex in $\mathbb{Z}^s$, and integral intervals $I_j = [a_j, b_j] \cap \mathbb{Z}$ with $b_j > a_j$. Let

$$P = \bigcup_{j=0}^{s} (I_j \times \{q_j\}) \subset \mathbb{Z} \oplus \mathbb{Z}^s. \tag{2}$$

If all I_j have the same length after an integer shear, a known product theorem determines $Q(P)$. When the lengths differ, however, P is usually not lattice-affinely a product, and naive restriction of the periodic nerve to a transverse face misses certain horizontal simplices. The exact full-face lemma below resolves this issue and proves the same topological product classification for all such intervals.

The proof uses established methods—good-cover nerves, cellular gluing, and covering-space exact sequences, as in the methodological literature [5, 6, 9]—but its application requires an exact description of the finite-strand geometry. In particular we check the true facewise inverse image, rather than a smaller mixed-strand subcomplex. Similarly, the vector-partition methods behind our arithmetic example are classical [7]. The specific object $Q(P)$ should not be identified without proof with toric indispensable complexes [8].

2 Periodic nerve and lattice normalization

Throughout, $X \simeq Y$ signifies that there exist continuous maps $u : X \to Y$ and $v : Y \to X$ together with continuous one-parameter families of maps connecting $v \circ u$ to id_X and $u \circ v$ to id_Y. This convention fixes precisely what is meant by equivalence under continuous deformation; it is stronger than an equality of fundamental groups or of integral singular groups. Let L be a finitely generated free abelian group and $P \subset L$ a finite set spanning the lattice $\langle P - P \rangle_{\mathbb{Z}} = L$. We call P *lattice-convex* if $P = \mathrm{conv}(P) \cap L$. Define

$$\mathcal{N}(P) = \{F \subset L : F \text{ finite and } c - F \subseteq P \text{ for some } c \in L\}, \qquad Q(P) = |\mathcal{N}(P)|/L. \tag{3}$$

As usual the empty face is included. L acts on $\mathcal{N}(P)$ by translation. A nonzero lattice translation cannot preserve a finite simplex setwise, so the action is free on simplices and $Q(P)$ is a CW complex. A choice of $0 \in P$ leads to the known difference presentation

$$\pi_1 Q(P) = \left\langle X_p \ (p \in P),\ X_0 = 1 \;\middle|\; X_p^{-1} X_q = X_u^{-1} X_v \text{ if } q - p = v - u \right\rangle. \tag{4}$$

The displacement group map $\delta_P : \pi_1 Q(P) \to L$ takes X_p to p; the difference presentation and marking are established in [2, 4].

Lemma 2.1 (Lattice-convexity and alignment)**.** *The configuration P from (2) is lattice-convex and spans $\mathbb{Z}^{s+1}$. There is an affine unimodular change of coordinates mapping it to*

$$P' = \bigcup_{j=0}^{s} (\{0, 1, \ldots, m_j\} \times \{q_j\}), \qquad m_j = b_j - a_j \geq 1.$$

Proof. The projection of $\mathrm{conv}(P)$ to $\mathbb{R}^s$ is the unimodular simplex $\mathrm{conv}\{q_0, \ldots, q_s\}$. Its only integer points are the q_j, and the slice above each vertex is exactly $[a_j, b_j]$. Hence $\mathrm{conv}(P) \cap \mathbb{Z}^{s+1} = P$. A horizontal unit difference is present in each strand. Differences

between the strand starts, together with that horizontal vector, generate $\mathbb{Z}^{s+1}$, since the $q_j - q_0$ form a $\mathbb{Z}$-basis in the transverse component. Finally the integral shear

$$(t, y_1, \dots, y_s) \mapsto \left(t - a_0 - \sum_{j=1}^{s} (a_j - a_0) y_j, y_1, \dots, y_s \right)$$

has determinant one and aligns all left endpoints to zero. □

From now on assume all strands are aligned, put $m_j \geq 1$ and $M = \max_j m_j$, and write $T = \{q_0, \dots, q_s\} \subset \mathbb{Z}^s$. Let $K = \mathcal{N}(T)$, $H = \mathbb{Z}$ be the horizontal translation subgroup, and $G = \mathbb{Z}^s$ the transverse subgroup. Projection on the transverse coordinate is a simplicial map

$$p : \mathcal{N}(P) \longrightarrow K,$$

with compatible H- and G-translation actions.

3 The corrected full-face lemma

For a nonempty simplex σ of K, let X_σ be the *full induced subcomplex* of $\mathcal{N}(P)$ on all vertices whose transverse coordinate belongs to the vertex set of σ. This is the actual geometric inverse image of $|\sigma|$, not a choice of mixed simplices.

Write B_q^m for the horizontal simplicial complex on $\mathbb{Z} \times \{q\}$ whose simplices are the finite sets of horizontal diameter at most m. It is the union of the full translated horizontal simplices of width m.

Lemma 3.1 (Horizontal interval complexes)**.** *For any integer $m \geq 1$, B_q^m is contractible. The quotient $|B_q^m|/H$ is a $K(\mathbb{Z}, 1)$, and the horizontal barycenter modulo $\mathbb{Z}$ is an equivalence under continuous deformation with S^1.*

Proof. Cover B_q^m by full simplices supported on the intervals $[h - m, h] \cap \mathbb{Z}$, $h \in \mathbb{Z}$. Every nonempty finite intersection is a simplex. Their abstract cover nerve consists of finite nonempty $D \subset \mathbb{Z}$ with $\max D - \min D \leq m$. This nerve is also the nerve of the intervals $[h - m, h] \subset \mathbb{R}$, which cover the real line and have contractible intersections. The two applications of the nerve theorem show $|B_q^m| \simeq \mathbb{R}$, hence it is contractible. Horizontal H acts freely; the quotient has universal cover B_q^m, so is a $K(\mathbb{Z}, 1)$. Barycenter changes by 1 under the deck generator and induces an isomorphism on π_1, giving the asserted equivalence under continuous deformation. □

Lemma 3.2 (Full transverse-face preimage)**.** *For every nonempty simplex $\sigma \subset K$ the full complex X_σ is contractible. If σ is a vertex $\{q\}$, then $X_\sigma = B_q^M$. If $|\sigma| \geq 2$ in vertices, there is a contractible mixed-row core C_σ and uniquely assigned indices $j(q) \in \{0, \dots, s\}$ such that*

$$X_\sigma = C_\sigma \cup \bigcup_{q \in \sigma} B_q^M, \qquad C_\sigma \cap B_q^M = B_q^{m_{j(q)}}. \tag{5}$$

The distinct B_q^M are disjoint, and all inclusions in (5) *are cellular cofibrations.*

Proof. For a singleton transverse vertex, that vertex can assume every role among the translates of T; its horizontal simplices consequently arise from all strand widths, whose union is the width-M complex.

For clarity, the uniqueness claim used for mixed faces follows directly from $T = \{0, e_1, \dots, e_s\}$: every nonzero difference $q_i - q_j$ uniquely determines the ordered pair (i, j). Indeed, vectors of the forms $\pm e_i$ and $e_i - e_j$ are distinct unless their ordered indices agree (the assertion

also holds for $s = 1$). Thus two different transverse translates of T cannot contain the same ordered pair of distinct vertices. A mixed transverse simplex has one role assignment, whereas a singleton has several.

Suppose σ contains at least two vertices. A nonzero difference $q_i - q_j$ with $q_i, q_j \in T$ uniquely specifies the ordered pair (i, j): the simplex vertices $\{0, e_1, \dots, e_s\}$ are affinely independent. Therefore σ belongs to a unique maximal translate of the transverse simplex K and assigns a unique row role $j(q)$ to each $q \in \sigma$. Let C_σ be the subcomplex generated by the mixed-row full simplices

$$D_h(\sigma) = \bigcup_{q \in \sigma} (([h - m_{j(q)}, h] \cap \mathbb{Z}) \times \{q\}), \quad h \in \mathbb{Z}.$$

Their finite intersections are simplices and are nonempty exactly when the horizontal centers have diameter at most $M_\sigma = \max_{q \in \sigma} m_{j(q)} \geq 1$. The same interval-nerve argument as in Lemma 3.1 gives $C_\sigma \simeq *$.

Every face involving at least two distinct transverse vertices belongs to this core by the uniqueness of its transverse maximal translate. A face involving just one transverse vertex may instead be witnessed by another translate of T and need not belong to the mixed core: it lies precisely in the full-width horizontal complex B_q^M. These two cases exhaust the actual simplices of X_σ, proving the union identity. Restricting a core simplex to the strand at q gives horizontal diameter at most $m_{j(q)}$, and every such horizontal simplex is present in the core; thus the stated intersection is exact.

By Lemma 3.1, C_σ, $B_q^{m_{j(q)}}$, and B_q^M are all contractible. A subcomplex inclusion $B_q^{m_{j(q)}} \hookrightarrow B_q^M$ is therefore an equivalence under continuous deformation and a cofibration. Attaching the width-M strand along that inclusion preserves the deformation class of the core by the CW gluing theorem for equivalences along cofibrations. Repeating for the finitely many $q \in \sigma$ proves $X_\sigma \simeq *$. □

Remark 3.3 (Why the full preimage matters)**.** *The tempting identification $X_\sigma = C_\sigma$ is false for unequal intervals. For example, take $s = 2$, strand widths $(m_0, m_1, m_2) = (1, 1, 4)$ and transverse face $\{0, -e_1\}$. Its mixed core uses the first two short strands, yet the horizontal edge with endpoints $(0, 0)$ and $(4, 0)$ is a genuine simplex of X_σ witnessed by the third, longer role. Formula (5) accounts for that edge. This repair is essential for the proof of Theorem 4.1.*

4 Product rigidity and marked displacement

Theorem 4.1 (Simplex-strand product rigidity)**.** *For every $s \geq 1$ and every configuration P of the form (2) with all $|I_j| \geq 2$,*

$$Q(P) \simeq Q(T) \times S^1 \simeq \left(\bigvee_{j=1}^{s} S^1 \right) \times S^1. \tag{6}$$

The displacement-marked fundamental group is $(F_s \times \mathbb{Z}, \mathrm{ab})$, up to the intrinsic integral basis of the active lattice.

Proof. After Lemma 2.1, write $L = H \oplus G$, with $H = \mathbb{Z}$ horizontal and $G = \mathbb{Z}^s$ transverse, and set

$$Y = |\mathcal{N}(P)|/H.$$

The affine horizontal barycentric coordinate η on every simplex obeys $\eta(hx) = \eta(x) + h$ and is fixed under transverse translations. Consequently the map

$$f = (p, \eta \bmod \mathbb{Z}) : Y \longrightarrow |K| \times S^1$$

is continuous and strictly G-equivariant, with G translating the first factor.

For a transverse face σ, put $Y_\sigma = |X_\sigma|/H$. By Lemma 3.2, X_σ is contractible. Horizontal translation acts freely on it, so Y_σ is a $K(\mathbb{Z}, 1)$. Since a width-one horizontal edge occurs in every X_σ, $\eta \bmod \mathbb{Z}$ maps the deck generator to winding number one. Therefore $\eta : Y_\sigma \to S^1$ is an equivalence under continuous deformation, and hence so is

$$f|_{Y_\sigma} : Y_\sigma \longrightarrow |\sigma| \times S^1.$$

Importantly, these maps are restrictions of the *same* global f; they commute strictly with face inclusions, without any independently chosen facewise contractions.

For a finite transverse subcomplex $K' \subset K$, write $Y[K'] = p^{-1}(|K'|) \subset Y$. The preimage of $|\sigma|$ for any face σ is exactly Y_σ; since it is the full geometric preimage, inverse images of two faces intersect over $\sigma \cap \tau$. When a maximal simplex σ is attached to a finite subcomplex containing its entire boundary, the two squares

$$\begin{array}{ccc} Y[\partial\sigma] & \longrightarrow & Y_\sigma \\ \downarrow & & \downarrow \\ Y[K'_{\mathrm{old}}] & \longrightarrow & Y[K'_{\mathrm{old}} \cup \sigma] \end{array} \qquad \begin{array}{ccc} |\partial\sigma| \times S^1 & \longrightarrow & |\sigma| \times S^1 \\ \downarrow & & \downarrow \\ |K'_{\mathrm{old}}| \times S^1 & \longrightarrow & |K'_{\mathrm{old}} \cup \sigma| \times S^1 \end{array}$$

are cellular pushouts, with horizontal and vertical attaching maps subcomplex inclusions and hence cofibrations. The natural map f induces a map of squares. On the full simplex and, by induction, on the boundary and the previously attached part it is a weak equivalence. The CW gluing principle for weak equivalences makes it so on the new union. Induction over faces of finite K', followed by exhaustion by finite subcomplexes (compact maps from spheres have images in a finite CW subcomplex), gives a weak equivalence

$$f : Y \xrightarrow{\simeq} |K| \times S^1.$$

We still need to descend this statement to the actual periodic quotient. The transverse complex K is locally finite because T is finite. No nonzero element of G preserves a simplex of K, so the translation action of G on $|K|$ is free and properly discontinuous: each point has a neighborhood U whose distinct G-translates are disjoint. Equivariance of $p : Y \to |K|$ gives the same property for $p^{-1}(U)$ in Y. Thus both quotient projections are regular covering maps with deck group G, not merely free cellular quotients. The map f induces the identity on that deck group. Both covering spaces are connected: K is connected because the edge differences of the unimodular simplex generate G, and Y is connected by the horizontal edges and the mixed simplices. Their regular covering sequences therefore give a commuting diagram

$$\begin{array}{ccccccccc} 1 & \to & \pi_1(Y) & \to & \pi_1(Y/G) & \to & G & \to & 1 \\ & & \downarrow\cong & & \downarrow & & \| & & \\ 1 & \to & \pi_1(|K| \times S^1) & \to & \pi_1((|K| \times S^1)/G) & \to & G & \to & 1. \end{array}$$

The middle arrow is an isomorphism by a group-extension diagram chase; the higher groups π_j also agree, because covering spaces preserve them for $j \geq 2$. The quotient map is therefore a weak equivalence and hence, by the CW Whitehead theorem, an equivalence under continuous deformation:

$$Q(P) = Y/G \simeq (|K|/G) \times S^1 = Q(T) \times S^1.$$

For the primitive simplex T, every maximal simplex of K is a translate of $-T$. Two different translates cannot share an edge: equality of a nonzero ordered edge difference determines its unique ordered pair of vertices of T, and hence determines the translation. More generally, no two distinct non-vertex faces of $-T$ lie in the same translation orbit. Consequently $Q(T)$ is the quotient of one geometric s-simplex by identifying its $s+1$ vertices and making no further identifications. Let A denote this finite vertex subcomplex. Since $A \hookrightarrow \Delta^s$ is a CW cofibration and Δ^s is contractible, its cofiber Δ^s/A has the homotopy type of the unreduced suspension of A, which is a wedge of s circles. Thus $Q(T) \simeq \bigvee^s S^1$; this also recovers the primitive-simplex case of [3].

Finally, the displacement homomorphism $\delta_P : \pi_1 Q(P) \to L$ is surjective: the differences $P - P$ generate the full active lattice L, and the difference presentation (4) realizes these differences as displacements of loops. By the product equivalence just proved, $H_1(Q(P);\mathbb{Z}) \cong \mathbb{Z}^{s+1}$, while $L \cong \mathbb{Z}^{s+1}$. Because L is abelian, δ_P factors through a surjection between free abelian groups of the same finite rank; that map is an isomorphism. Hence the marking is exactly abelianization, up to the intrinsic unimodular choice of coordinates on L. □

Corollary 4.2 (Integral invariants)**.** *In the setting of Theorem 4.1,*

$$\pi_1 Q(P) \cong F_s \times \mathbb{Z}, \quad H_0(Q(P);\mathbb{Z}) \cong \mathbb{Z}, \quad H_1(Q(P);\mathbb{Z}) \cong \mathbb{Z}^{s+1}, \quad H_2(Q(P);\mathbb{Z}) \cong \mathbb{Z}^s,$$

$H_j(Q(P);\mathbb{Z}) = 0$ *for* $j \geq 3$*, and* $\pi_j Q(P) = 0$ *for all* $j \geq 2$*. In particular* $Q(P)$ *is aspherical and has no torsion in its integral singular groups. The linear displacement kernel* $\ker(H_1 \to L)$ *vanishes; the full zero-displacement subgroup is* $[F_s, F_s] \times \{0\}$*.*

Proof. A wedge of circles is a graph and hence aspherical. Its product with S^1 is aspherical. Integral Künneth gives the stated integral singular groups, and the marking follows from Theorem 4.1. □

5 An exact shortest-completion example with a topological transition

Consider the fixed integer source with six genuine unit-cost actions

$$A = \begin{pmatrix} -1 & 2 & 1 & 0 & 3 & -2 \\ -2 & 3 & 0 & 1 & 2 & -1 \end{pmatrix}. \tag{7}$$

Its image is $\mathbb{Z}^2$, since columns $(1,0)$ and $(0,1)$ occur; the strictly positive vector $(2,1,1,1,1,2)$ lies in $\ker_{\mathbb{Z}} A$. Consequently every integer residual has a nonnegative source completion: for an integral preimage u one may add a large positive multiple of the strictly positive kernel vector to make u nonnegative. The source is a genuine rank-two counting system, not a globally recognized one-counter language.

For target $r = (x, y)$ set $S = x + y$ and $D = x - y$. The six action vectors in (D, S) coordinates are precisely $\{\pm 1\} \times \{-3, 1, 5\}$.

Proposition 5.1 (Exact length and counting profile)**.** *For any* $r = (x, y) \in \mathbb{Z}^2$ *and* $N \geq 0$*, a source Parikh vector* $z \in \mathbb{N}^6$ *with* $Az = r$ *and* $|z| = N$ *exists exactly when*

$$N \geq |D|, \qquad -3N \leq S \leq 5N, \qquad N \equiv S \pmod 4. \tag{8}$$

When this holds, put $P = (N + D)/2$*,* $R = (N - D)/2$*,* $K = (S + 3N)/4$ *and*

$$F_j(t) = \sum_{a_0 + a_1 + a_2 = j} t^{a_1 + 2a_2} = \begin{bmatrix} j+2 \\ 2 \end{bmatrix}_t.$$

Then the number of Parikh vectors *of length N at r equals*

$$c_A(r, N) = [t^K]F_P(t)F_R(t). \tag{9}$$

Thus $L_A(r)$ is the least integer $N \geq \max\{|D|, S/5, -S/3, 0\}$ satisfying $N \equiv S \pmod 4$.

Proof. Every action has $D = \pm 1$ and $S = -3+4j$, $j = 0, 1, 2$, proving necessity. Conversely, (8) makes P, R nonnegative integral, with $P + R = N$, and makes $0 \leq K \leq 2N$ integral. Decompose K into N digits from $\{0, 1, 2\}$ and assign P of them to the $D = +1$ columns and the remaining R to $D = -1$ columns; each sign group offers all three digit choices. This produces the required vector. At fixed P, R the choices of multiplicities of each digit in each sign group are independent, hence the generating-function product and coefficient formula. This counts unordered source-letter multiplicities, not word permutations. □

Let $r_N = (N-1, 1)$ for $N \geq 3$. Then $S = N$, $D = N-2$, and (8) gives $L_A(r_N) = N$. The length-N fiber has one $D = -1$ letter and $N-1$ $D = +1$ letters. Index the chosen negative letter by digit $j \in \{0, 1, 2\}$ and put $t = z_5$, the multiplicity of the original fifth action. Solving the remaining equations gives

$$z_1 = t + j - 1, \qquad z_3 = N - j - 2t, \qquad z_5 = t,$$

with exactly one of z_6, z_4, z_2 equal to one, corresponding to $j = 0, 1, 2$ respectively. Its integral affine fiber coordinates are

row j	allowed t
0	$1 \leq t \leq \lfloor N/2 \rfloor$
1	$0 \leq t \leq \lfloor (N-1)/2 \rfloor$
2	$0 \leq t \leq \lfloor (N-2)/2 \rfloor$.

(10)

This bijection preserves the full active difference lattice, and the total number of optimal vectors is $\lfloor 3N/2 \rfloor$. For $N = 3$ the row sizes are $(1, 2, 1)$. The *entire* shortest Parikh fiber, in the original six-action coordinates, consists of

$$(0,0,1,0,1,1), \qquad (0,0,2,1,0,0),$$
$$(1,0,0,1,1,0), \qquad (1,1,1,0,0,0).$$

Using the first point as origin, the three difference vectors have a 3×3 minor of determinant $+1$ on original coordinates $1, 2, 3$. They therefore form an active *primitive* three-simplex (the full active lattice has rank three). For every $N \geq 4$, each of the three complete rows in (10) contains at least two consecutive points and their transverse coordinates form a primitive triangle.

Corollary 5.2 (Target-indexed topological transition)**.** *For the* same fixed source (7)*, the true full shortest-completion fibers satisfy*

$$Q(P_A(r_N)) \simeq \begin{cases} \bigvee^3 S^1, & N = 3, \\ (S^1 \vee S^1) \times S^1, & N \geq 4. \end{cases}$$

Thus H_2 changes from 0 at $N = 3$ to $\mathbb{Z}^2$ for all $N \geq 4$, even though active rank is three throughout and the optimal Parikh multiplicity grows as $\lfloor 3N/2 \rfloor$.

Proof. The $N = 3$ case is the primitive-minimal-fiber theorem. For $N \geq 4$ apply Theorem 4.1 to the three true rows (10). The integral-group conclusion follows from Corollary 4.2. □

6 Toric-fiber interpretation and theorem hypotheses

Every signed integer counting source can be embedded into a pointed nonnegative semigroup by adjoining a degree coordinate. Given $A = (a_1, \ldots, a_n)$, choose $c \in \mathbb{Z}^d$ with $a_i + c \geq 0$ for every i, and form B_c with columns $(a_i + c, 1)$. The tautological identity

$$\{z \in \mathbb{N}^n : Az = r,\ |z| = N\} = \{z \in \mathbb{N}^n : B_c z = (r + Nc, N)\}$$

identifies fixed-length signed Parikh sets with degree-graded pointed toric fibers. The associated scalar counts belong to established vector partition theory [7]; in our example the Gaussian coefficient expression is an explicit specialization of standard fiber-counting methods. Conversely, our periodic nerve $Q(P)$ is built from lattice translations in the *full* minimum-grade fiber and is not, by definition, the indispensable complex attached to a toric ideal [8].

Theorem 4.1 is stated for unimodular transverse simplices and intervals containing at least two consecutive lattice points. The target-indexed formula in Corollary 5.2 concerns the minimum-grade fibers of the fixed counting matrix (7); the input source, rather than its parameters, is held constant. The result is an intrinsic statement about the periodic quotient of these exact Parikh fibers, without requiring a physical pushdown realization.

7 Conclusion

The exact full-face decomposition repairs a subtle but consequential pitfall in applying functorial nerve methods to unequal-width periodic strands: simplices supported on a single transverse vertex can arise from roles outside a chosen mixed transverse simplex. Accounting for these additional horizontal thickenings makes the face preimages contractible and natural, allowing the deformation equivalence to descend through both translation groups. The resulting rigidity theorem computes a complete aspherical deformation class, its integral singular groups, and its displacement marking for a uniform class of non-product optimal-fiber configurations. A fixed rank-two completion-total source then supplies a target-indexed topological transition followed by stability under unbounded optimal multiplicity.

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